

Apply Filters to SQL Queries

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Scenario

This scenario is based on a fictional company as part of Coursera's Google Cybersecurity Professional Certificate.

Your cybersecurity team needs data to investigate potential security issues and to update computers. You are tasked with obtaining specific information about employees, their machines, and the departments they belong to from the database. All of the data is to be accessed using MariaDB.

My tasks:

Utilize MariaDB to perform filtering commands (AND, OR, NOT) for your team:

- 1. Retrieve after hours failed login attempts*
- 2. Retrieve login attempts on specific dates*
- 3. Retrieve login attempts outside of Mexico*
- 4. Retrieve employees in Marketing*
- 5. Retrieve employees in Finance or Sales*
- 6. Retrieve all employees not in IT*

SQL Queries

Retrieve after hours failed login attempts

Queried the log_in_attempts table to identify failed login attempts that occurred after 18:00 (after hours). Using the AND operator allowed me to filter for records where both the success status was false (0) and the login time was outside standard working hours.

Command:

```
SELECT * FROM log_in_attempts WHERE success = 0 AND login_time > '18:00:00';
```

Retrieve login attempts on specific dates

Queried the log_in_attempts table to isolate login activity that occurred on specific dates using the OR operator. This allowed me to retrieve records matching either May 9, 2022, or May 10, 2022, in order to analyze event patterns across targeted timeframe windows.

Command:

```
SELECT * FROM log_in_attempts WHERE login_date = '2022-05-09' OR login_date = '2022-05-10';
```

Retrieve login attempts outside of Mexico

Queried the log_in_attempts table to identify activity originating outside of Mexico using the NOT operator combined with LIKE 'MEX%'. This allowed me to filter out any records associated with Mexico or its country codes to focus exclusively on international login traffic.

Command:

```
SELECT * FROM log_in_attempts WHERE country NOT LIKE 'MEX%';
```

Retrieve employees in Marketing

Queried the employees table to filter for records where the department column is equal to 'Marketing'. Using the WHERE clause with the equals operator allowed me to isolate and retrieve information specifically for employees belonging to the Marketing team.

Command:

```
SELECT * FROM employees WHERE department = 'Marketing';
```

Retrieve employees in Finance or Sales

Queried the employees table to filter for staff members in either the Finance or Sales departments using the OR operator. This enabled me to retrieve records matching either condition in a single query.

Command:

```
SELECT * FROM employees WHERE department = 'Finance' OR department = 'Sales';
```

Retrieve all employees not in IT

Queried the employees table to isolate team members outside of the Information Technology department using the NOT operator. This allowed me to filter out all records where the department equals 'IT' and return employees from all other departments.

Command:

```
SELECT * FROM employees WHERE department != 'IT';
```

Summary

In this project, I used MariaDB to apply conditional SQL filters to retrieve specific security and employee data from organizational databases. By using filtering operators such as AND, OR,

and NOT, I successfully isolated failed after-hours login attempts, international traffic outside of Mexico, and specific departmental records. Practicing these queries demonstrated how SQL filters streamline data investigation for cybersecurity tasks, allowing security teams to efficiently analyze log events and manage system access.